Airline Customer Insights

Surveys, complaints and passenger data: what drives satisfaction and loyalty

Raisa AK | Python · PostgreSQL · Power BI | Synthetic practice dataset

HEADLINE

Refunds, check-in and Business Class loyalty are the biggest gaps

- Overall: CSAT 69.8%, NPS +5.1, CES 3.59, FCR 63.0%, 1.00 complaints per 1,000 passengers.
- Business Class NPS is −6.5, far below First (+27.3) and Economy (+7.7), despite similar CSAT.
- Refund (63.9%) and Check-in (64.2%) have the lowest CSAT of all issue types.
- JFK has the most complaints per 1,000 passengers (1.57); CDG has the lowest CSAT (57.4%) and NPS (−5.1).
- Lounge Access complaints are solved at first contact only 50% of the time.

DATA & APPROACH

Cleaned 3 tables, logged every fix, built one trusted model

Data: 500 surveys, 400 complaint cases, 240 route-cabin-month passenger rows (Jan–Oct 2024).

Pipeline:

1. Python (Pandas): clean + data-quality log
2. PostgreSQL: star schema (Routes dimension, PK/FK) + KPI views
3. Power BI: Overview, Complaints, Drivers pages

Cleaning rules: dedupe on record IDs (not customer); invalid values set to null, never guessed; every fix logged.

Issue	Rows	Action
Duplicate Survey/Case IDs	5	Removed
Scores outside valid range	7	Set to null
Cabin / issue / resolved spellings	14	Standardised
Routes with extra spaces	6	Trimmed
Contact count 0 or negative	4	Set to null
Negative passengers	2	Set to null
Passengers blank or text	5	Kept as null

WHAT IS HAPPENING

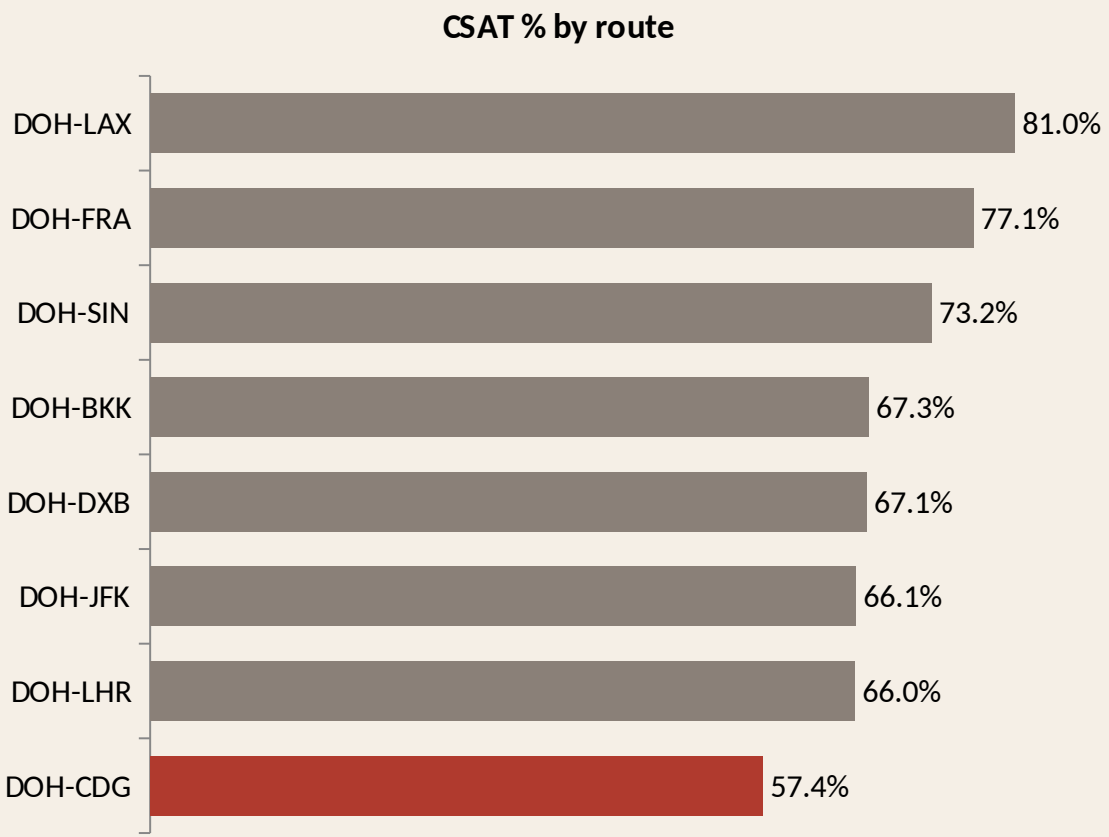
Satisfaction is moderate, loyalty is only just positive

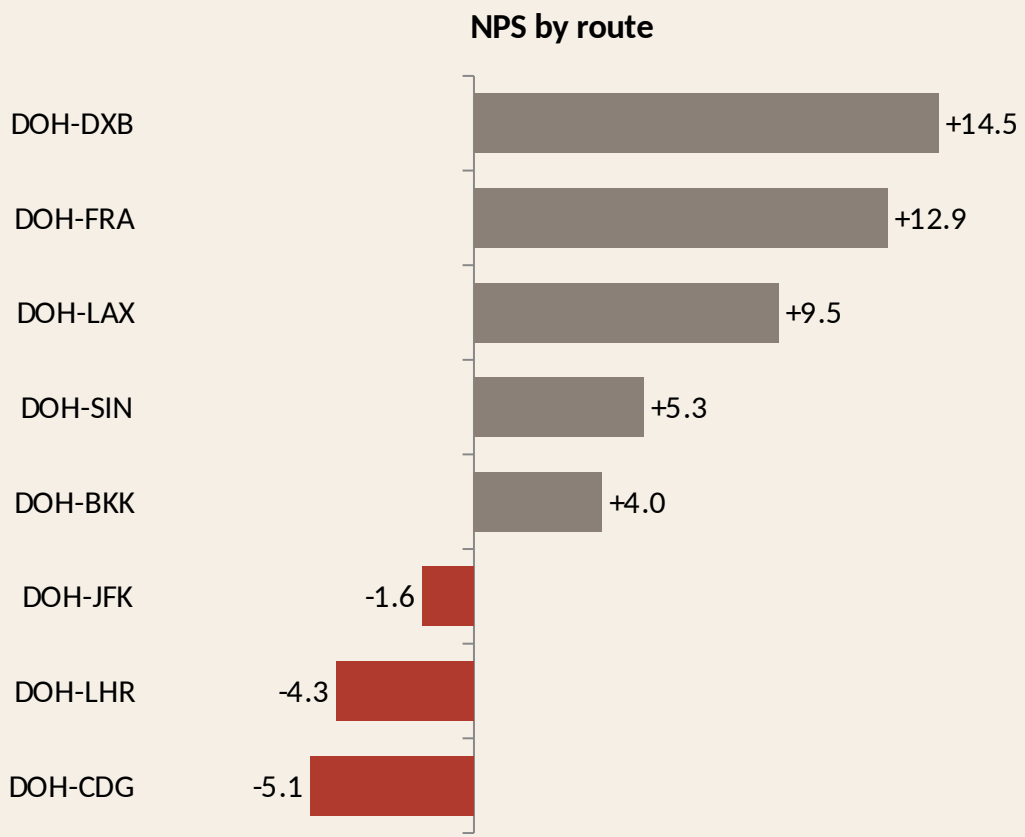
69.8%	+5.1	3.59	63.0%	1.00
CSAT	**NPS**	**CES**	**FCR**	**Complaints**
scores 4–5	promoters – detractors	1–7, lower is better	solved in 1 contact	per 1,000 pax

Roughly 7 in 10 passengers are satisfied, but promoters only slightly outnumber detractors, so satisfaction is not yet turning into loyalty.

WHY: ROUTES

CDG and LHR trail on both satisfaction and loyalty

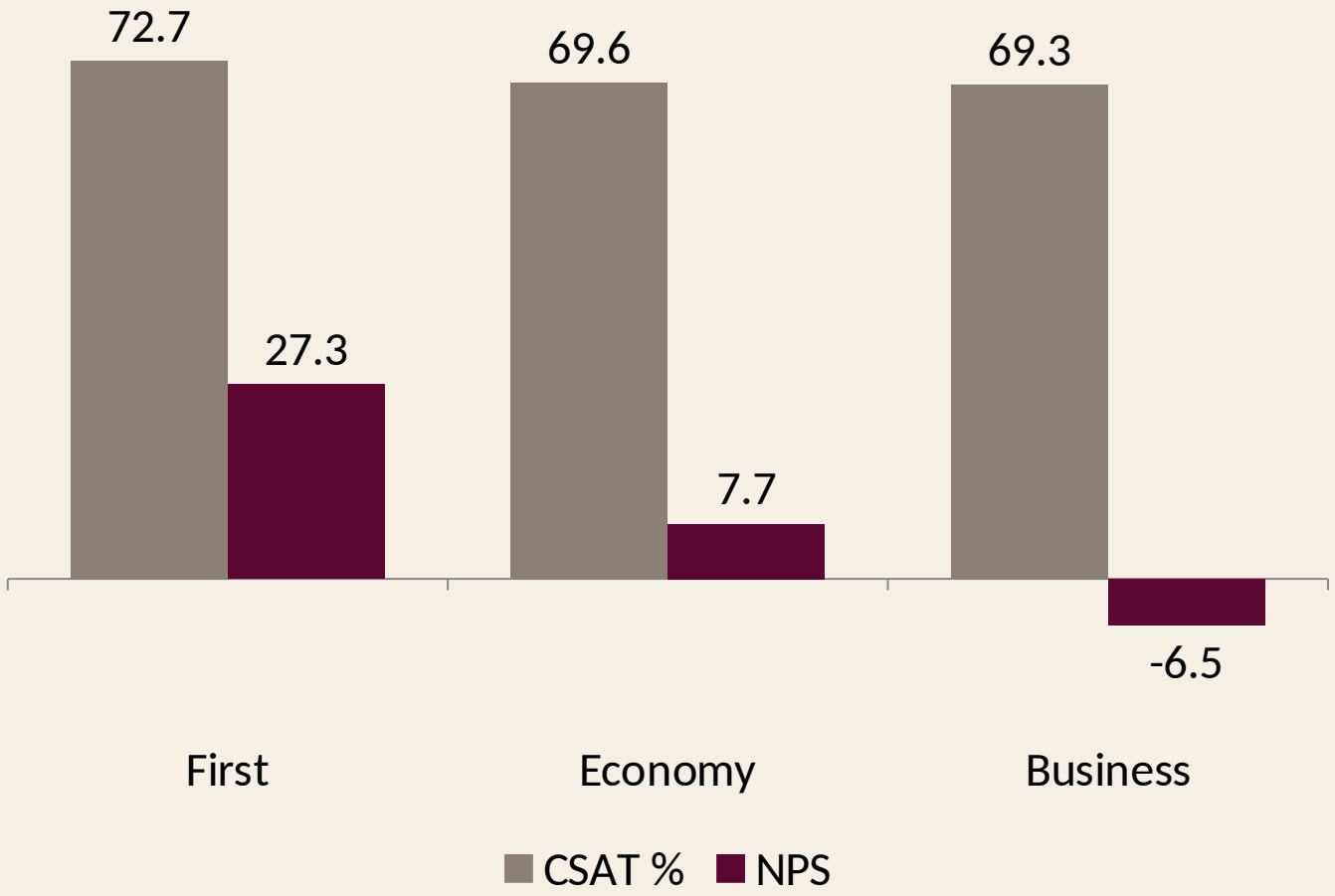




DOH-CDG is the weakest route: lowest CSAT (57.4%) and negative NPS (−5.1). LHR and JFK also have more detractors than promoters.

WHY: CABIN CLASS

Business Class passengers are satisfied but not loyal



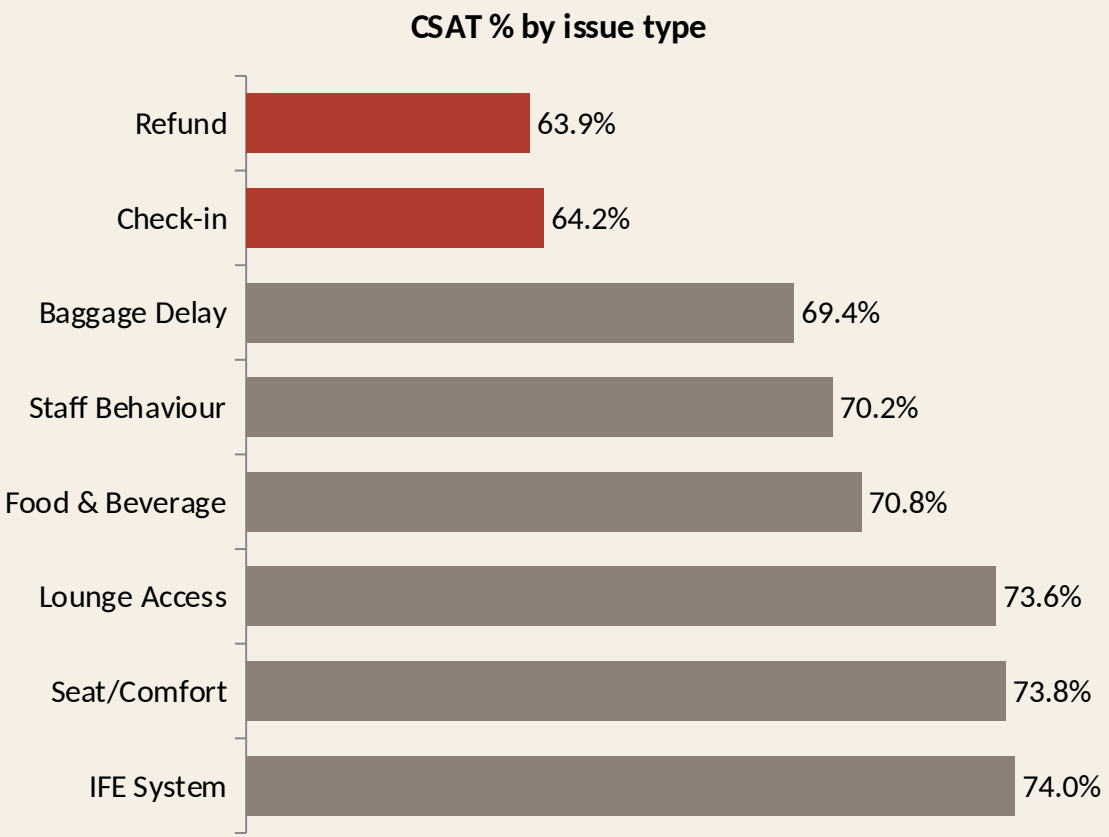
CSAT is almost the same in every cabin (69–73%).

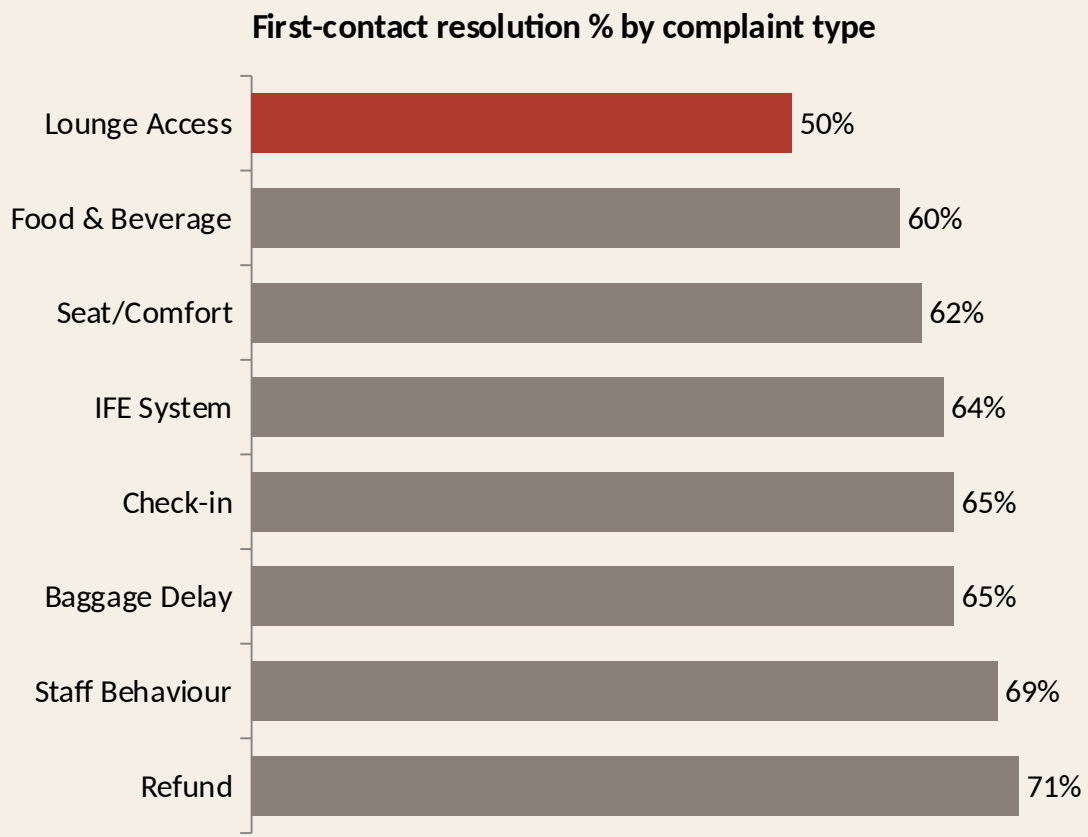
But NPS splits sharply: First +27.3, Economy +7.7, Business −6.5.

Business passengers rate single touchpoints well but would not recommend us. This is a loyalty risk for high-value customers. Next step: read Business comments and compare with competitors.

WHY: TOUCHPOINTS

Refunds and check-in hurt satisfaction; lounge cases are hard to resolve

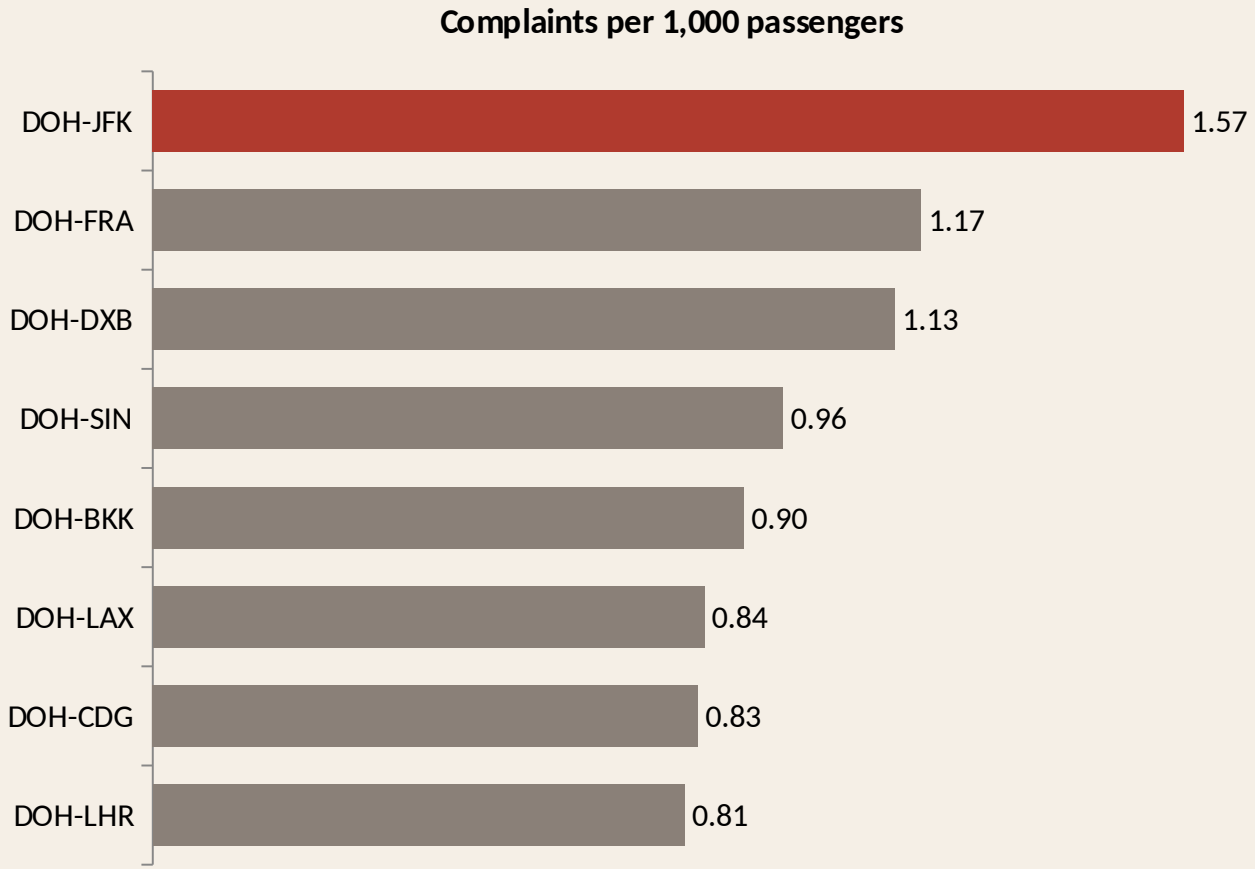




Lounge Access has the most complaint cases (60) and the lowest FCR (50%): half need a second contact.

WHY: COMPLAINT RATE

JFK generates the most complaints for its passenger volume



Raw counts would mislead: routes with more passengers naturally get more complaints.

Normalised per 1,000 passengers, JFK (1.57) is nearly double LHR (0.81).

115 customers raised more than one complaint, a clear service-recovery opportunity.

RECOMMENDATIONS

Three actions, each with a KPI to track

1. Fix the refund and check-in experience

Simplify refund tracking and set a refund SLA; review check-in queues at weakest stations.

Track: CSAT % for Refund and Check-in, avg resolution days

2. Win back Business Class loyalty

Deep-dive Business comments on CDG, LHR and JFK; benchmark seat, lounge and service vs competitors.

Track: Business NPS by route

3. Resolve lounge cases first time

Give agents lounge-access rules and authority to resolve on first contact; proactively contact repeat complainers.

Track: Lounge FCR %, repeat complainers

NEXT STEPS & LIMITS

What I would do next

- Test significance before acting: chi-square for CSAT % gaps, t-test for CES averages. Some route-cabin groups have fewer than 30 responses, so treat those as directional.
- Add a shared Date table to compare months (MoM) and track the impact of fixes.
- Analyse open-text comments to explain the Business Class NPS gap.
- Fix data quality at the source with IT: dropdowns for cabin and issue type, validation on score ranges, required fields, and a data dictionary.
- Note: synthetic practice dataset; numbers illustrate the method, not real airline performance.